Healthy Skepticism Session 2

Fault 2 Probility Reversal

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[venue and date]

The Two Questions

Two probabilities that sound the same and are not

WHAT CAN BE MEASURED

P(observation | condition)

When the condition is there, how often do we see this?

WHAT YOU WANT TO KNOW

P(condition | observation)

Having seen it, how likely is the condition?

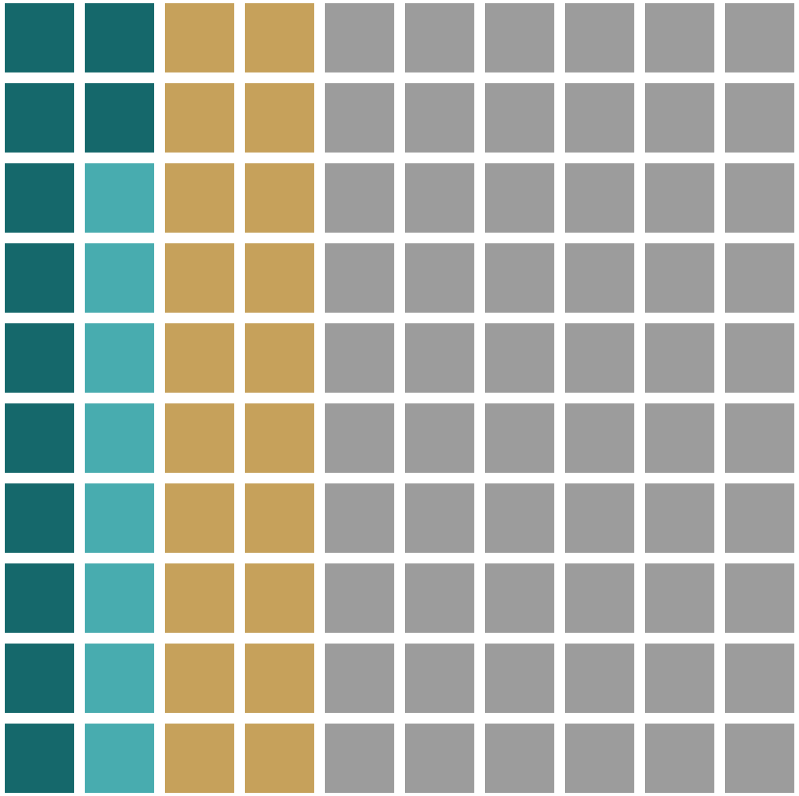
The fault is reading one as if it were the other. They are different numbers, and how far apart they sit depends on something the observation never tells you: how common the condition is.

EACH SQUARE IS A MEMBER OF THE POPULATION

Healthy Skepticism

How to read a waffle chart

One square, one member. Call the two attributes **A** and **B**.



Two attributes, four groups

	Both **A** and **B**	**12%**	of the population
	A only	**8%**	of the population
	B only	**20%**	of the population
	Neither **A** nor **B**	**60%**	of the population

Writing it as a probability

	P(A and B)	=	**12%**
	P(A only)	=	**8%**
	P(B only)	=	**20%**
	P(neither)	=	**60%**

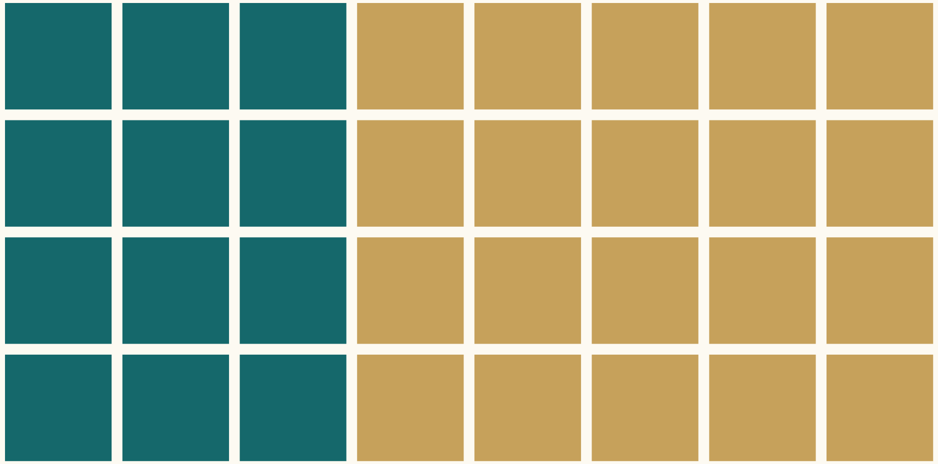
P(...) is just a name for a percentage of the grid. The four add to 100%: every square is counted once.

RESTRICTION OF ATTENTION, NOT SHRINKAGE

What P(A | B) Actually Means

P(A | B) reads “A given B.” Because we are **assuming B is true**, the question becomes: of the B population, what percentage are A?

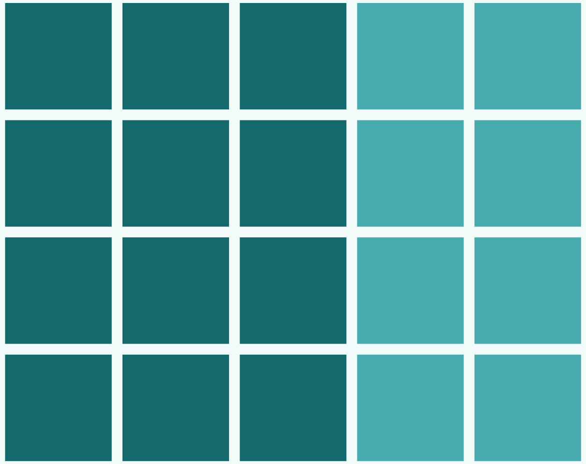
Given B — attend only to the 32 B’s



P(A | B) = 12 of 32 = 37.5%

Denominator = 32, the size of the B population.

Given A — attend only to the 20 A’s



P(B | A) = 12 of 20 = 60%

Denominator = 20, the size of the A population.

Dark teal = in both A and B · Gold = B only · Light teal = A only. *Same 12 in the middle, two different answers.*

A TEST RESULT TELLS YOU WHICH POPULATION YOU ARE NOW STANDING IN

Test Results Are Examples of Conditional Probability

Let A = “the test says positive” and B = “the person actually has the condition.” Sensitivity and predictive value are two different conditionals — and they are not close.

Prevalence P(B) = 3% • Sensitivity P(A | B) = 96% • Specificity P(not A | not B) = 95% • 10,000 people

	Has condition	No condition	Total
Test +	288	485	773
Test –	12	9,215	9,227
Total	**300**	**9,700**	**10,000**

Read down the “Has condition” column for sensitivity: 288 of 300.
Read across the “Test +” row for predictive value: 288 of 773.

Given B — restrict to the 300 who have it

P(A | B) = 288 / 300 = 96%

Sensitivity — a property of the test

Given A — restrict to the 773 who test positive

P(B | A) = 288 / 773 = 37%

Positive predictive value — what the patient asks

More than half of all positive results are false alarms: 485 of 773.

A positive result sends you for a second, more specific test — not to treatment.

THE TWO EXAMPLES

Two Real World Examples

A SCREENING MAMMOGRAM

P(test + | has cancer) is high

sensitivity — the number the test is sold on

BUT THE REVERSED CONDITIONAL

P(has cancer | test +) is low — the false positives outnumber the real cancers.

A COVID RAPID TEST

P(test – | not infected) is high

specificity — the number the test is sold on

BUT THE REVERSED CONDITIONAL

P(not infected | test −) is not high enough — false negatives walk out infectious.

Both tests report P(result | condition). What you need is P(condition | result) — and reversing it fails in opposite directions: the mammogram floods you with false positives, the rapid test sends you home with a false negative.

THE FIRST REVERSAL

A Screening Mammogram

If it is positive, do I have cancer?

A REAL TEST, REAL NUMBERS

What the mammogram is measured on

SENSITIVITY

P(called back | cancer)

87 of every 100 women with cancer

the other 13 are missed

A positive mammogram is a **callback**. The radiologist wants more pictures. It is not a diagnosis, and nothing has been found yet.

This is the number the mammogram is judged on, and it is a good one. But notice where it starts: from already knowing she has cancer. That is the only direction a test can be measured in.

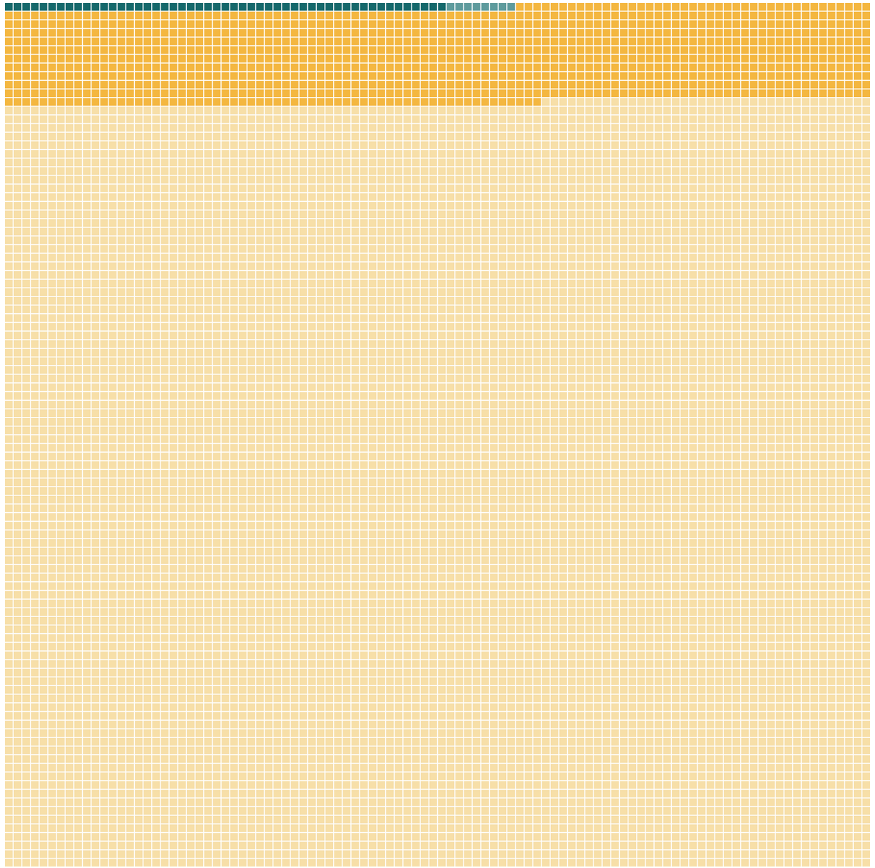
What we want is P(cancer | called back): 4.4%

Same test, same day. Now we walk the reversal.

Sensitivity 86.9%. Lehman CD et al. National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 2007-2013.

AN ILLUSTRATIVE COHORT OF 10,000 WOMEN

Six in a thousand have it



Every square is one woman screened once. The teal sliver at the top is everyone who has cancer.

That is an average across ages 40 to 89. It climbs with age: nearer 4 per 1,000 in the fifties, 6 or 7 in the sixties.

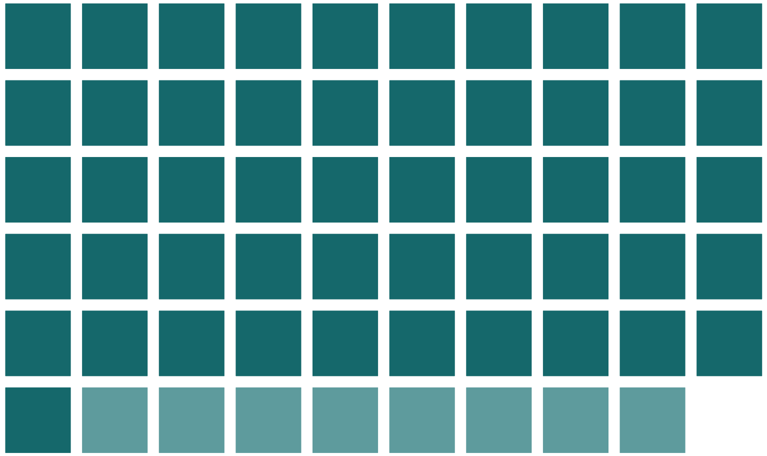
- has cancer, called back: **51**
- has cancer, not called back: **8**
- no cancer, called back anyway: **1,103**
- no cancer, all clear: **8,838**

The amber band is eleven rows deep. It is more than twenty times the size of the group the test is looking for.

BCSC benchmarks applied to a 10,000-woman cohort. Lehman CD et al. National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 359 radiologists, 95 facilities, 2007-2013.

FIRST QUESTION

If you have cancer, how likely is a callback?



The 59 women who have cancer, enlarged from the 10,000.

Look only at the 59 women who have cancer. Everyone else is set aside for now.

Of those 59, the ones called back: **51**

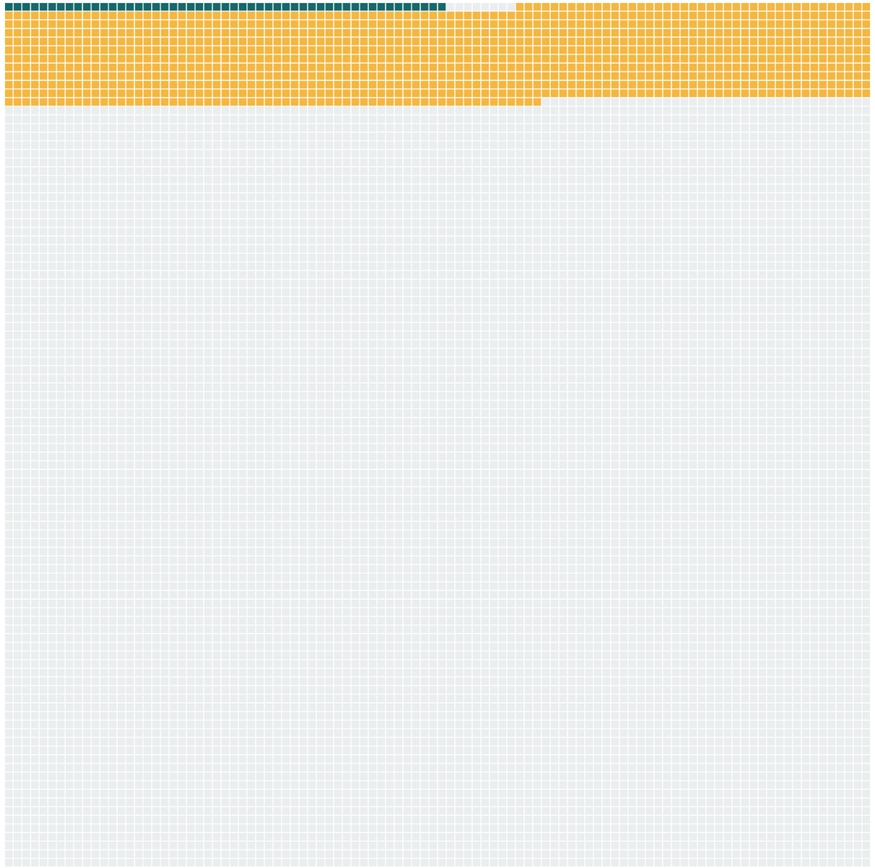
P(callback | cancer) = 51 / 59 = 87%

This is the number the mammogram is judged on, and it is a good one.

Sensitivity 86.9%. Lehman CD et al. National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 359 radiologists, 95 facilities, 2007-2013.

SECOND QUESTION, THE ONE PEOPLE GET BACKWARDS

If you get a callback, how likely is cancer?



Now look only at the women who get called back. Same picture, we just divide by a different group.

Of those 1,154, the ones who have cancer: **51**

P(cancer | callback) = 51 / 1,154 = 4.4%

Twenty-two of every twenty-three callbacks are false alarms. This is the value the benchmark study itself publishes.

Positive predictive value of an abnormal interpretation, 4.4%. Lehman CD et al. National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 359 radiologists, 95 facilities, 2007-2013.

THE REVERSAL, FOR ONE WOMAN

Unlikely, but not ignorable

$$P(\text{ called back } \mid \text{ cancer }) \;=\; 87\%$$

↓ *reversed becomes*

$$P(\text{ cancer } \mid \text{ called back }) \;=\; 4\%$$

87% is what the mammogram was measured on. 4% is what her letter means. Same test, same day, same woman.

Unlikely is why she should not panic. Not ignorable is why the biopsy is warranted.

Positive predictive value of an abnormal interpretation 4.4%, of a biopsy recommendation 25.6%. Lehman CD et al., Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58.

THE BIGGER PICTURE

Why reasonable people disagree

Of US women aged 50, screened every year for ten years:

avoid a breast cancer death	**0.03% to 0.32%**
have at least one false alarm	**49% to 67%**
are overdiagnosed, and treated for it	**0.3% to 1.4%**

Nobody in this argument disputes the sensitivity, the specificity, or the 4.4%. The disagreement is over how to weigh a false alarm for half of them against a death averted for a fraction of one percent.

So the guidance splits. The American College of Radiology recommends annual screening from 40. The Task Force says every two years from 40. The Nordic Cochrane group and the Swiss Medical Board have argued for not running screening programmes at all.

Ranges from published estimates for women aged 50 screened annually for a decade. Overdiagnosis is contested: the Canadian National Breast Screening Study's 25-year follow-up put 106 of 484 screen-detected cancers (21.9%) as overdiagnoses; Gotzsche and Jorgensen, Cochrane 2013 (CD001877), found no effect on overall mortality across ten trials. An 11.6% callback rate compounded over ten independent screens gives about 71%, near the top of the false-alarm range.

THE SECOND REVERSAL

A COVID Rapid Test

If it is negative, am I clear?

A REAL TEST, REAL NUMBERS

What the swab is measured on

SPECIFICITY

P(negative | no COVID)

99 of every 100 people who are well

almost nobody healthy is wrongly flagged

That is about as specific as any test in common use. It is why a positive swab is worth believing.

But look where it starts: from already knowing the person is well. Nobody handed a negative result knows that.

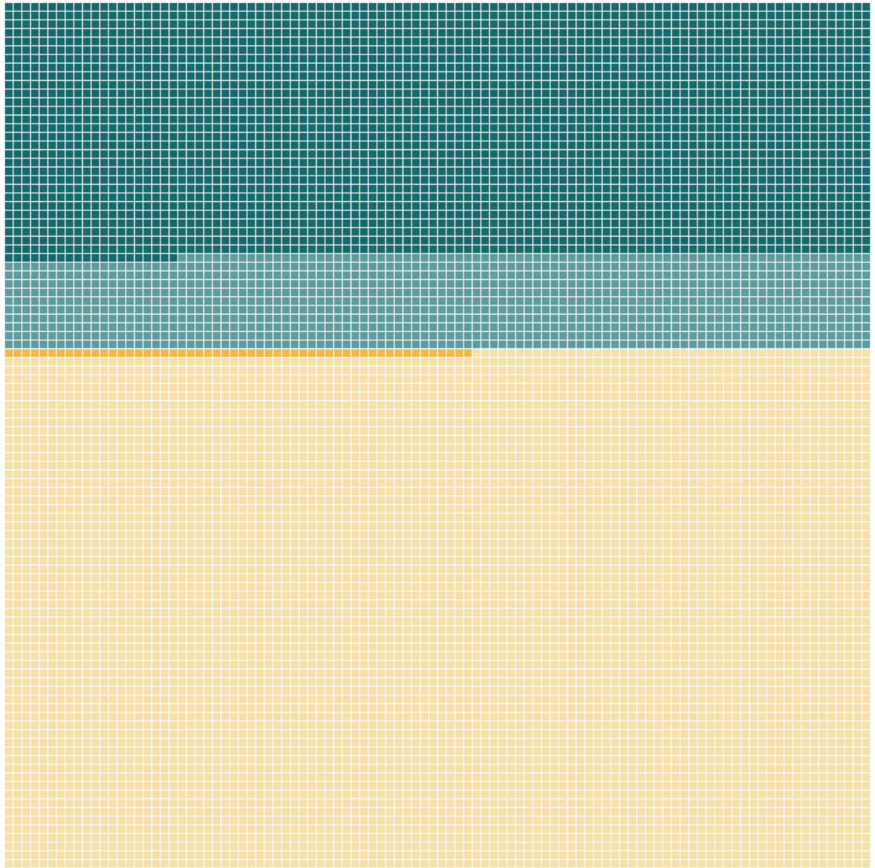
What we want is P(no COVID | negative): 85%

Nowhere near certainty. Now we walk the reversal.

Rapid antigen tests in symptomatic people: specificity 99.1%, sensitivity 73.0% (95% CI 69.3 to 76.4). Dinnes et al., Cochrane Database of Systematic Reviews 2022;7:CD013705.

AN ILLUSTRATIVE COHORT OF 10,000 PEOPLE WITH SYMPTOMS

Four in ten of them have it



Everyone here feels ill during a wave, so the odds before testing are far higher than in the street outside.

- has COVID, tests positive: **2,920**
- has COVID, tests negative: **1,080**
- no COVID, tests positive: **54**
- no COVID, tests negative: **5,946**

The pale teal band is the one to watch. Eleven rows of people who have COVID and were told they did not.

Pre-test probability of 4 in 10 is illustrative, chosen to represent symptomatic testing during a wave. Test figures: Rapid antigen tests in symptomatic people: sensitivity 73.0% (95% CI 69.3 to 76.4), specificity 99.1%. Dinnes et al., Cochrane Database of Systematic Reviews 2022;7:CD013705.

THE REVERSAL, FOR ONE PERSON

A negative is not an all-clear

$$P(\text{ negative} \mid \text{no COVID }) = 99\%$$

↓ *reversed becomes*

$$P(\text{ no COVID} \mid \text{negative }) = 85\%$$

99% is what the swab was measured on. 85% is what your negative means.

More than one person in seven who was told they were negative had COVID and went out anyway. A negative narrows the odds. It does not clear you.

Counts carried from the previous slide: 7,026 negatives, of which 1,080 had COVID. Rapid antigen tests in symptomatic people: specificity 99.1%, sensitivity 73.0% (95% CI 69.3 to 76.4). Dinnes et al., Cochrane Database of Systematic Reviews 2022;7:CD013705.

WHEN NOBODY CATCHES IT

Courtrooms and Classrooms

What did the flip cost?

A CAUTIONARY TALE

When the reversal reaches a verdict

A RAPE TRIAL IN ENGLAND, CONVICTION QUASHED 1994

The forensic expert told the court that the likelihood of the sample coming from any man other than the accused was one in three million.

That number describes how rare the profile is. It was offered as the chance he was innocent, the judge repeated it to the jury, and the Court of Appeal quashed the conviction.

TRAFFIC STOPS IN HOUSTON, 2004 TO 2015

A two-dollar roadside kit turns blue for cocaine. It also turns blue for dozens of other things, which has been known since the 1970s.

Blue was taken to mean cocaine. The lab checked only afterwards, and found more than 250 people had already pleaded guilty to possessing nothing.

Both times a number about how often this happens to the innocent was read as the chance of innocence.

R v Deen, Court of Appeal, reported The Times 10 January 1994; rules on presenting DNA statistics were later set in Doheny and Adams [1997] 1 Cr App R 369. Houston: Houston Forensic Science Center and the Harris County conviction integrity unit, reported by ProPublica and The New York Times Magazine, 2016.

A THIRD CAUTIONARY TALE

The detector that cannot tell

WHAT THE VENDORS ADVERTISE

Turnitin has claimed 98%. Copyleaks 99.1%. GPTZero 99%. Winston AI 99.98%.

Every one of those is a statement about machines: if this text came from an AI, we will flag it.

WHAT A FLAGGED STUDENT NEEDS

Of the essays you flag, how many were actually written by an AI?

Vanderbilt ran that number. Even at a 1% false positive rate, their 75,000 submissions a year would falsely accuse 750 students. They switched the tool off in August 2023.

And 1% is generous. Stanford researchers ran seven detectors over 91 TOEFL essays, every one written by a human: 61.3% were flagged as machine-written, and about one in five was flagged by all seven. On essays by native English speakers the same detectors almost never erred.

The same flip, with a transcript at the end of it. A number about how often the tool catches machines, read as the chance that a student cheated.

Liang W, Yuksekgonul M, Mao Y, Wu E, Zou J. GPT detectors are biased against non-native English writers. Patterns 2023. Vanderbilt University disabled Turnitin's AI detection in August 2023. Johns Hopkins and international-student false positives reported by The Markup, 14 August 2023. Vendor accuracy figures are the vendors' own marketing claims.

What we’ve learned

One way a true number can mislead you, and a look ahead.

1 · Probability reversal

The trap: What the test reports gets flipped into what you asked. A positive is read as proof, a negative as the all-clear.

Ask: out of everyone who got this result, how many actually have it?

2 · Next session: Arbitrary categories

The trap: A line drawn across a continuous measurement is read as a fact about your body.

Ask: where do I sit on the curve, not which side of the line?

Questions, and a conversation

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